

SENIOR ANALYTICS CASEBOOK

Decisions built on defensible evidence

Three end-to-end public-data case studies spanning consumer risk, healthcare quality, and labor economics.

113,689 source rows profiled	3 federal agencies	100% reproducible pipeline
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What this portfolio proves

A senior analyst does more than make charts: the work starts with a decision, establishes data contracts, quantifies uncertainty, tests assumptions, and ends with an operational recommendation.

42,426 Arizona consumer complaints analyzed	5,419 U.S. hospitals quality-profiled	1939-2026 employment history covered	0.23% seasonal-naive forecast MAPE
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Senior analyst operating model

Stage	Evidence standard
01 · Frame	Translate ambiguity into a decision, accountable owner, grain, outcome metric, and guardrail.
02 · Govern	Document provenance, validate keys, surface missingness, and preserve reproducible transformations.
03 · Analyze	Combine descriptive, diagnostic, segmented, and predictive views; stress-test the result.
04 · Activate	Prioritize actions by expected impact, implementation effort, monitoring signal, and risk.

Portfolio architecture

Raw official extracts → schema and quality checks → analysis-ready models → validated metrics → executive narrative → downloadable evidence. The same metric definitions appear in the site, casebook, CSV outputs, and reference SQL.

The analyses are deliberately cross-domain. That breadth demonstrates transferable judgment: issue taxonomy and concentration risk; missingness-aware quality screening; time-series normalization, shock recovery, and backtesting.

Data as retrieved in September 2026. Public records may be revised by source agencies; results are portfolio demonstrations, not regulatory, clinical, or investment advice.

Data quality is part of the answer

A polished output built on weak contracts is not senior analysis. Each dataset was checked at the decision-relevant grain before metrics were produced.

Dataset	Validated key	Raw rows	Key uniqueness	Key metric coverage	Primary constraint
CFPB complaints	Complaint ID	42,426	100.00%	100.00%	State = AZ; dates = calendar 2024
CMS hospitals	Facility ID	5,419	100.00%	58.57%	Ratings missing by design for many facilities
BLS monthly series	Series + month	65,844	100.00%	100.00%*	Annual M13 rows excluded; series aligned by month

Six-dimensional quality framework

Dimension	Control
Completeness	Required-field and analytic-variable coverage; missingness shown explicitly.
Uniqueness	Business keys tested before any aggregation or join.
Validity	Dates, enumerations, and numerical ranges checked against source definitions.
Consistency	Comparable labels standardized without erasing source provenance.
Timeliness	Extract dates and revision risk documented next to the result.
Traceability	Published metrics map back to code, processed tables, and official sources.

Important nuance

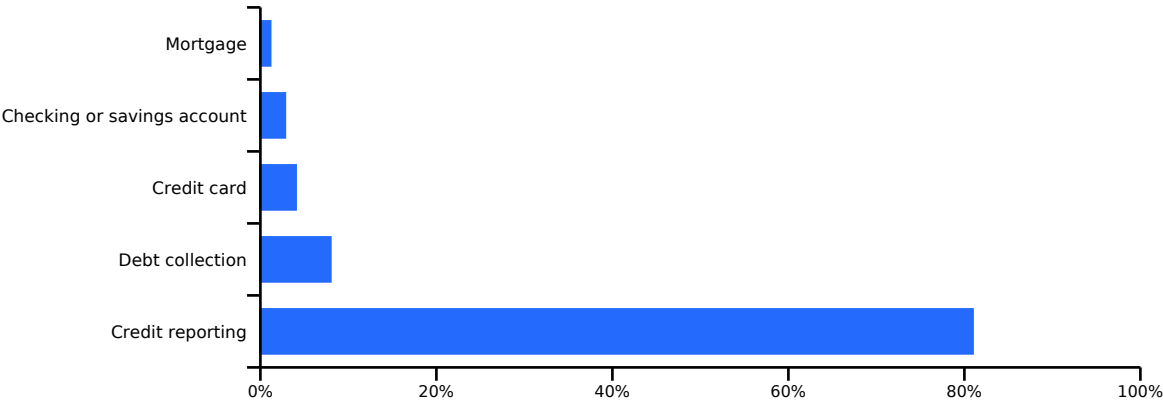
CMS star-rating coverage is 58.57%. Excluding unrated facilities would make the rated subset look complete, while treating missing ratings as poor performance would invent evidence. This portfolio reports the denominator, evaluates coverage by segment, and keeps selection logic auditable.

*BLS analytical coverage refers to the joined monthly range used for each metric; source series begin on different dates. No duplicate series-month rows were found.

Complaint intelligence for Arizona

Decision: where should a consumer-operations leader focus root-cause work, partner oversight, and monitoring capacity?

42,426 complaints received in 2024	80.9% from credit-reporting products	83.0% at the five largest companies	99.68% company timely-response rate
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Finding

The issue mix is highly concentrated: credit-reporting complaints account for 80.9% of the sample. Company concentration is also material—the five largest recipients account for 83.0%. Timely response is nearly universal, so response speed alone is a weak differentiator; leaders should shift attention to recurrence, correction quality, and preventability.

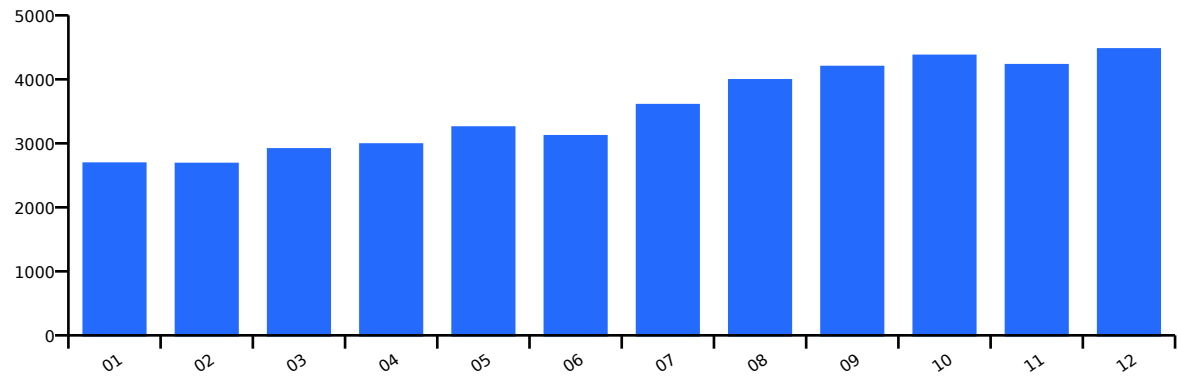
Recommended action

Stand up a weekly credit-reporting defect review, beginning with the three nationwide bureaus. Track repeat issue, product, company, and monthly rate—not only case volume. Add a control chart when exposure denominators become available.

Source: Consumer Financial Protection Bureau, Consumer Complaint Database; Arizona complaints received 2024-01-01 through 2024-12-31. Company responses reflect source fields and are not independent adjudications of wrongdoing.

From volume to an operating queue

The analytical move is from a static Pareto chart to a recurring management cadence with explicit escalation rules.



December was the peak month

Complaint volume reached 4,467 in 2024-12, versus 2,682 in January. That pattern justifies a monthly alert but does not establish causation. The next test should normalize by consumers or accounts and distinguish source growth from underlying service deterioration.

Horizon	Workstream	Deliverable
0-30 days	Taxonomy	Map top issues and sub-issues to process owners; audit duplicate-like submissions.
31-60 days	Root cause	Sample top-company narratives and dispositions; separate data furnishing from dispute handling.
61-90 days	Control	Publish recurrence, reopening, correction, and aging KPIs with thresholds.

Metric contract

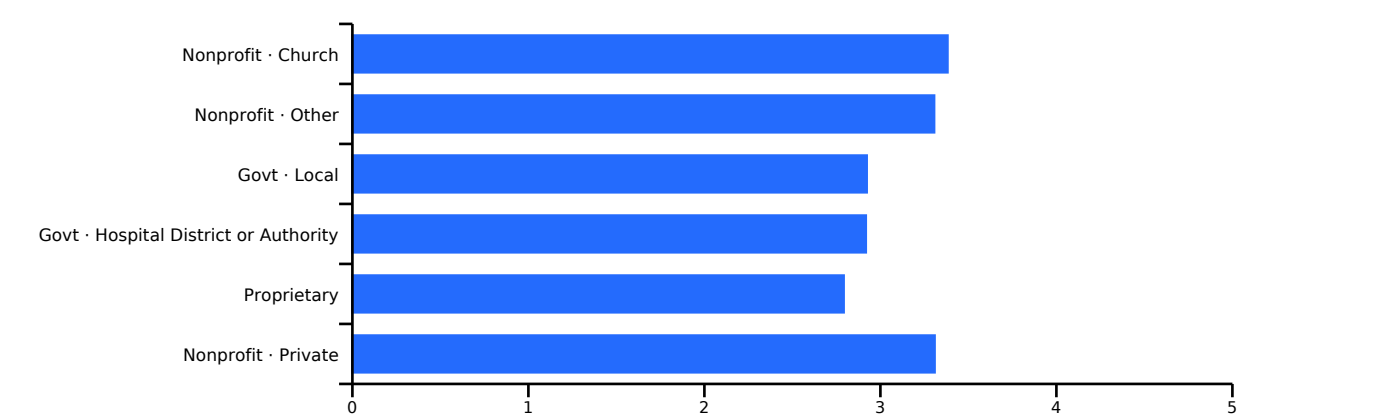
Complaint share = complaints in segment ÷ all filtered complaints. Grain = complaint ID. Window = received date in calendar 2024. Geography = Arizona. Guardrails = do not infer consumer harm severity from volume; do not compare organizations without exposure denominators.

Sensitivity and limitations: complaint behavior varies by consumer awareness and channel; database text may contain unverified allegations; product labels can change over time; Arizona was selected to create a bounded state-level operating example.

A transparent hospital screening model

Decision: how should a network, employer, or operations team create a defensible shortlist without hiding missing ratings?

5,419 facilities profiled nationally	58.6% with an overall star rating	1,179 high-quality screen matches	3.11 Arizona rated-facility average
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Screening rule

A facility qualifies when it has an overall rating of 4 or 5 and is not flagged as worse than the national value on the selected safety indicator. The rule produced 1,179 candidates. This is a prioritization screen—not a clinical ranking—and should be followed by service-line, access, and local validation.

Interpretation

Average rating varies by ownership category, but the comparison is descriptive. Coverage also differs by ownership and facility type, so raw means should not be read as causal performance effects.

Source: Centers for Medicare & Medicaid Services, Hospital General Information, release dated 2026-08-13. Overall ratings and safety fields are defined by CMS; portfolio screening logic is the analyst's own.

Use the shortlist as a starting point

A senior workflow separates eligibility, ranking, and final selection—and exposes each tradeoff to the decision owner.

State	Hospitals	Rating coverage	Average rating	High-quality share
VA	96	79.2%	3.68	42.7%
UT	52	55.8%	4.24	42.3%
WI	140	61.4%	3.78	37.9%
PA	186	70.4%	3.48	34.4%
CT	36	72.2%	3.42	33.3%
NH	27	85.2%	3.17	33.3%
SC	66	74.2%	3.43	31.8%

Governance-ready selection framework

Layer	Control
Eligibility	Meets service-line, geography, credentialing, and data-availability requirements.
Quality screen	Overall stars ≥ 4 and no selected worse-than-national safety flag.
Access layer	Travel time, capacity, appointment lag, language access, and network status.
Equity check	Compare exclusion rates and data coverage across communities and facility types.
Review cadence	Quarterly source refresh; exception review when CMS revises methodology.

Do not collapse every signal into one opaque score

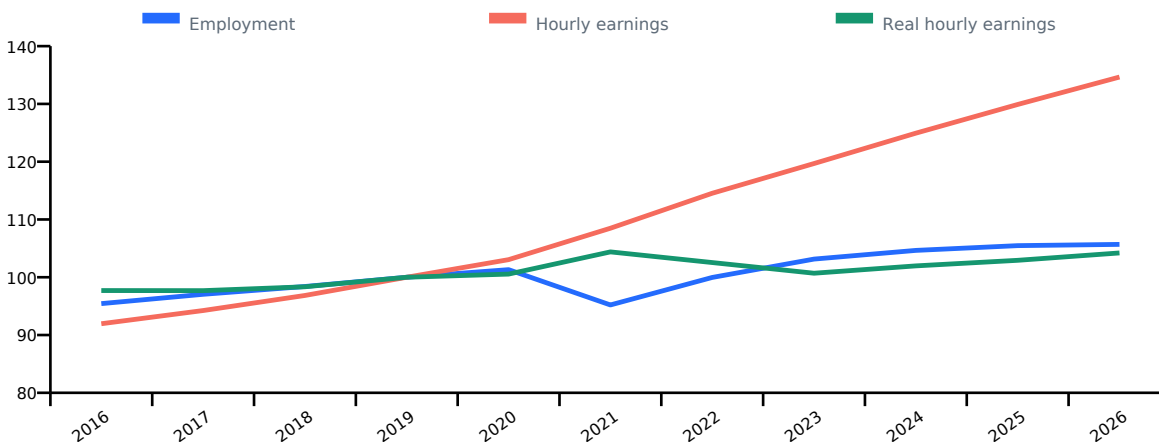
Weighted composite scores imply tradeoffs that stakeholders may not have authorized. Preserve the component measures, show how candidate counts change under alternative thresholds, and require human review for borderline or missing-data cases.

Limitations: overall ratings are broad summaries; missing ratings are not random; facilities can differ in case mix, services, reporting eligibility, and measurement reliability. The shortlist is unsuitable for individual medical decisions.

Separate jobs, pay, and purchasing power

Decision: what should an executive workforce brief say when nominal pay rises but inflation changes the lived outcome?

159.1M latest total nonfarm employment	+603.0K 12-month employment change	102.81 real wage index, Jan 2019 = 100	Jun 2022 employment recovered pre-shock level
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Finding

Employment was 159.1 million in August 2026, up 603,000 year over year (+0.38%). The pandemic shock produced a 14.36% one-month drop, with the pre-shock employment level recovered in June 2022. Indexing wages and CPI to the same base makes the distinction between nominal and real gains visible.

Executive interpretation

A hiring plan should pair labor availability with real compensation pressure. Nominal wage growth alone can overstate employee purchasing-power gains; headcount growth alone can mask a slowing market.

Sources: U.S. Bureau of Labor Statistics, Current Employment Statistics series for total nonfarm employment and private-sector hours/earnings; Consumer Price Index for All Urban Consumers. Values may be revised by BLS.

Backtest before you forecast

Complexity is not a credential. The model that best survives an honest holdout is the stronger baseline.

0.23% seasonal-naïve holdout MAPE	0.93% linear-trend holdout MAPE	4.0× relative error of trend vs. seasonal baseline
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Model	Definition	Holdout MAPE	Decision
Seasonal naïve	Value from the same month one year earlier	0.23%	Selected baseline
Linear trend	OLS on the preceding monthly history	0.93%	Rejected for this holdout

What I would add in production

Control	Production requirement
Revisions	Vintage the monthly extracts and score forecasts against first-release and revised actuals.
Intervals	Publish prediction intervals and scenario bands, not only point estimates.
Drift	Monitor error, residual autocorrelation, breakpoints, and changes in seasonal structure.
Decision tie-in	Map forecast scenarios to recruiting capacity, workforce budget, and trigger thresholds.

Why the simpler model won

On the documented holdout, seasonality captured the short-horizon level more reliably than an unrestricted trend. Model selection follows observed generalization error, not prestige. A new candidate must beat this baseline out of sample and remain explainable to the decision owner.

Forecast validation is illustrative and uses published historical observations. Results are sensitive to the holdout window and future revisions; they are not economic or investment forecasts.

Auditable work, ready to discuss

The portfolio package includes the source-aware pipeline, processed analytical tables, reference SQL, methodology, and this executive casebook.

Capability	Evidence in this portfolio
Python / pandas	Cleaning, validation, segmentation, joins, descriptive statistics, rank correlations, and backtests.
SQL	CTEs, window functions, Pareto concentration, missingness-aware aggregation, and time-series feature logic.
Data visualization	Decision-first chart selection, consistent metric contracts, accessible labels, and restrained color.
Analytical judgment	Causal restraint, denominator discipline, uncertainty disclosure, sensitivity checks, and action design.

Source register

Agency	Dataset	Official source
CFPB	Consumer Complaint Database	consumerfinance.gov/data-research/consumer-complaints
CMS	Hospital General Information	data.cms.gov/provider-data/dataset/xubh-q36u
BLS	CES + CPI public time series	bls.gov/ces and bls.gov/cpi

LET'S TALK

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